ByteXL-MRECW AIstack-GenAI -AgentiAI-IoT Innovation Readiness Training (38 Days)

Days 1-12: Foundation and Core Learning

Day	Topic	Description	Key Points	Mapped Topics
Day 1	Overview on Python Programming & Web Technologies	Establish core programming skills with Python and version control with Git. Introduce the fundamental building blocks of web development, including structuring content with HTML, styling with CSS, and adding interactivity with JavaScript.	Python syntax, data types, control flow, functions, OOP basics, Git commands, HTML structure, CSS styling, DOM manipulation, basic JavaScript logic.	Programming & Web Foundations
Day 2	Data Science Essentials	Focus on data manipulation, analysis, and visualization using Python's powerful libraries. Understand how to prepare and explore data, which is crucial for any AI/ML application.	Pandas DataFrames, NumPy arrays, data wrangling, EDA, data visualization (Matplotlib, Seaborn).	Data Science Essentials
Day 3	Machine Learning	Introduce the core concepts of traditional machine learning, including supervised learning algorithms, model training, and evaluation metrics.	Supervised learning, classification, regression, model evaluation (accuracy, precision, recall, F1-score, R-squared, MSE), cross-validation, feature scaling.	Machine Learning
Day 4	Deep Learning & Neural Networks	Cover the basics of neural networks, their architecture, and the training process. Focus on fundamental concepts without advanced CNN architectures.	Perceptrons, multi-layer perceptrons, activation functions, forward/backward propagation, loss functions, optimizers, basic neural network concepts.	Deep Learning & Neural Networks
Day 5	Natural Language Processing (NLP) & LLMs	Explore how AI processes human language, from basic text analysis to the architecture and applications of Large Language Models.	Text preprocessing, embeddings, sentiment analysis, Transformer architecture, attention mechanism, pre-training vs. fine-tuning LLMs, Q&A systems.	Natural Language Processing (NLP) & LLMs
Day 6	Generative AI & Multimodal AI	Introduce the concepts behind generative models for creating new content and how AI systems integrate information from multiple modalities.	GANs, VAEs, Diffusion Models (conceptual), prompt engineering, synthetic data, data fusion, cross-modal learning, image captioning, video understanding.	Generative AI & Multimodal AI
Day 7	AI Agents & Agentic AI with LLM	Understand the principles of AI agents, their design, and how Large Language Models can power autonomous agents capable of planning and executing tasks.	Agent architectures, perception-action loop, task-oriented agents, multi-agent systems, autonomous agents, ReAct pattern, LLM planning, agent loops.	AI Agents & Agentic AI with LLM
Day 8	Cloud Computing & AI Deployment	Cover cloud platforms, AI model deployment, and scalable computing resources for AI applications.	Cloud platforms (AWS,**Azure,** GCP), containerization, model serving, API deployment, scaling strategies.	Cloud Computing & AI Deployment
Day 9	Data Engineering & Pipelines	Focus on building robust data pipelines, data processing workflows, and integration with various data sources.	ETL processes, data pipelines, batch vs. streaming processing, data quality, API integration.	Data Engineering & Pipelines

Day 10	Web APIs & Full-Stack AI Applications	Learn to develop web APIs for serving AI models and build interactive user interfaces using frameworks like Flask, Streamlit, and React for full-stack AI applications.	REST API principles, JSON handling, Flask routing, Streamlit dashboards, React components, full-stack integration, real-time dashboards with AI insights.	Web APIs & Full-Stack AI Applications
Day 11	MLOps, Databases & Security	Conclude with essential practices for managing the lifecycle of AI/ML models, storing data, and securing AI systems, along with final steps of project development and deployment.	SQL/NoSQL databases (MongoDB), MLOps (MLflow, DVC, CI/CD for ML), AI system security, authentication, project testing, cloud deployment, documentation.	MLOps, Databases & Security
Day 12	IoT Technology & Edge AI	Cover the basics of IoT hardware, embedded programming, and the deployment of AI models on edge devices, integrating with cloud platforms.	Microcontrollers (ESP32, Arduino), sensors, actuators, embedded programming, edge vs. cloud AI, TinyML, model optimization for edge.	
Days 13-38: Domain-Specific AI Projects (Cloud Deployment)				
Day 13-14	Project Implementation in Precision Agriculture Analytics	Develop an AI-driven agriculture system that analyzes crop data, predicts health issues, and provides farming recommendations through data analytics and machine learning.	Data pipeline setup, ML model development for crop prediction, image analysis for plant diseases, dashboard creation, database integration.	Data Science Essentials, AI Agents & Agentic AI with LLM, Machine Learning
Day 15-16	Project Implementation in Residential Energy Analytics	Design and implement an intelligent energy management system that analyzes household consumption patterns, forecasts usage, and provides optimization recommendations.	Energy data analysis, consumption forecasting, optimization algorithms, user interface design, data processing.	Machine Learning Fundamentals, Natural Language Processing (NLP) & LLMs, Data Engineering & Pipelines
Day 17-18	AI Integration in Smart Home Energy Analytics	Enhance the energy management system with AI-driven insights and automated recommendation systems.	AI model integration, automated insights, recommendation engines, data visualization.	Machine Learning Fundamentals, Natural Language Processing (NLP) & LLMs, Web APIs & Full-Stack AI Applications
Day 19-20	Intelligent System Design in Home Energy Optimization	Complete the energy optimization platform with advanced analytics and intelligent decision-making capabilities.	Advanced optimization algorithms, intelligent decision systems, comprehensive analytics dashboard.	Machine Learning Fundamentals, AI Agents & Agentic AI with LLM, Data Engineering & Pipelines
Day 21-22	Project Implementation in Smart Waste Management Analytics	Develop an AI-powered system that analyzes waste data, classifies waste types through image analysis, and optimizes collection strategies.	Image classification for waste types, optimization algorithms, data integration, analytics dashboard.	Programming & Web Foundations, Machine Learning Fundamentals, AI Agents & Agentic AI with LLM
Day 23-24	Project Implementation in EdTech and Adaptive Learning	Build an AI-driven educational platform that personalizes learning content and pathways, providing intelligent feedback and recommendations through data analysis.	Recommendation systems, NLP for feedback analysis, LLM integration for personalized learning, adaptive learning algorithms, user interface development.	Natural Language Processing (NLP) & LLMs, AI Agents & Agentic AI with LLM, Web APIs & Full-Stack AI Applications
Day 23-24	AI Project Implementation in Intelligent Supply Chain Analytics	Develop an AI-driven system that optimizes supply chain operations through data analytics, from inventory management to demand forecasting.	Demand forecasting models, inventory optimization algorithms, market analysis through NLP, supply chain analytics dashboard.	Machine Learning Fundamentals, Natural Language Processing (NLP) & LLMs, Data Science Essentials

Day 25-26	Solution Deployment in Digital Health Intelligence	Create an AI-driven platform that provides personalized health recommendations and assists in disease prediction through health data analysis.	Health data analytics, disease prediction models, personalized recommendation systems, secure data handling, LLM for health insights.	Machine Learning Fundamentals, Natural Language Processing (NLP) & LLMs, MLOps, Databases & Security
Day 27-28	Project Implementation in Personalized HealthTech Systems	Expand the health intelligence platform with advanced analytics and personalized care recommendations.	Advanced health analytics, personalized care algorithms, comprehensive health dashboards, patient data management.	Machine Learning Fundamentals, AI Agents & Agentic AI with LLM, Data Engineering & Pipelines
Day 29-30	AI Integration in Intelligent Urban Mobility Analytics	Develop an AI-driven system to analyze urban traffic patterns, predict congestion, and optimize transportation through data analytics.	Traffic pattern analysis, congestion prediction models, route optimization algorithms, urban mobility dashboard.	Machine Learning Fundamentals, Data Science Essentials, Web APIs & Full-Stack AI Applications
Day 33-34	Project Implementation in Environmental Data Analytics	Develop an AI-driven system to analyze environmental data, predict pollution patterns, and provide actionable insights for environmental protection.	Environmental data analysis, pollution prediction models, data integration and processing, environmental monitoring dashboards.	Machine Learning Fundamentals, Data Science Essentials, Data Engineering & Pipelines
Day 35-36	Project Development in Intelligent Retail Analytics	Develop an AI-driven system for retail analytics to analyze customer behavior, optimize business operations, and personalize customer experiences through data insights.	Customer behavior analytics, personalized recommendation systems, retail analytics, inventory optimization through data analysis.	Machine Learning Fundamentals, Natural Language Processing (NLP) & LLMs, Data Science Essentials
Day 37	AI System Implementation in Smart Energy Infrastructure Analytics	Develop an AI-driven system to analyze electricity distribution data, balance supply and demand through predictive analytics, and optimize energy grid operations.	Grid data analysis, energy forecasting models, optimization algorithms, energy analytics dashboard.	Machine Learning Fundamentals, Data Science Essentials, Web APIs & Full-Stack AI Applications
Day 38	AI Application Development in Emergency Response Analytics	Develop an AI-driven system to enhance disaster preparedness through predictive analytics, real-time data analysis, and intelligent resource coordination.	Disaster prediction models, real-time data analysis, resource optimization algorithms, emergency response analytics, crisis management dashboards.	Machine Learning Fundamentals, AI Agents & Agentic AI with LLM, Data Engineering & Pipelines